STAT 314500 - Applied Partial Differential Equations
Assignment 4: Laplace's Equation and Wave equation

This is an individual assignment. You may collaborate with others but are expected to submit your own work. Show all relevant work. Incomplete work will be docked points.

Note: if a question is marked with an asterisk, *, and appears vague - that's on purpose. Questions marked * are conceptual short-answer questions. Do your best to reason these through, and describe that reasoning concisely. We are looking more for reasoning than precision.

Reading: Chapters 2, 3 and 4 in Applied Partial Differential Equations with Fourier Series and Boundary Value Problems (Richard Haberman).

Question 1 Consider the Laplace's equation inside a rectangle $\Delta u = 0, (x, y) \in (0, L_1) \times (0, L_2)$ with boundary conditions

$$\partial_x u(0, y) = \partial_y(x, 0) = \partial_x u(L_1, y) = 0, \quad \partial_y u(x, L_2) = f(x)\,.$$

1. Briefly explain when there is a solution to this problem from a physical point of view. (2 points)
2. Solve this problem using separation of variables. Verify that the condition under which this works is only when part (i) is satisfied. (5 points)
3. The solution in part (ii) has an arbitrary constant. Determine this constant if u was the steady state solution to a heat equation $\partial_t v = k\Delta v$ with initial condition $g(x, y)$. (3 points)

Question 2 Solve the Laplace's equation outside a circular disk centred at the origin of radius R subject to the boundary condition

$$u(R, \theta) = \log 2 + 4\cos(3\theta)\,,$$

where (r, θ) denotes the polar coordinate representation. (7 points)

Question 3 Solve the Laplace's equation in a circular annulus $a < r < b$ subject to the boundary condition (7 points)

$$\partial_r u(a, \theta) = 0, \quad u(b, \theta) = g(\theta)\,.$$

Question 4 Consider the one dimensional wave equation

$$\begin{aligned}
\frac{\partial^2 u}{\partial t^2} &= c^2 \frac{\partial^2 u}{\partial x^2} \\
u(0, t) &= 0 \\
u(L, t) &= 0 \\
u(x, 0) &= f(x) \\
\frac{\partial u}{\partial x}(x, 0) &= g(x)
\end{aligned} \tag{1}$$

1. Separation of variables yields a solution of the form

$$u(x, t) = \sum_{n=1}^{\infty} A_n \cos\frac{n\pi x}{L} \sin\frac{n\pi ct}{L} + B_n \sin\frac{n\pi x}{L} \cos\frac{n\pi ct}{L}\,.$$

Show that this is equivalent to $u(x, t) = \phi(x + ct) + \psi(x - ct)$ for some functions ϕ and ψ. (5 points)

2. Define the energy

$$E(t) = \frac{1}{2}\int_0^L \left|\frac{\partial u}{\partial t}\right|^2 \mathrm{d}x + \frac{c^2}{2}\int_0^L \left|\frac{\partial u}{\partial x}\right|^2 \mathrm{d}x\,.$$

This can be thought of as the sum of a kinetic energy and a potential energy. Show that (7 points)

$$\frac{\mathrm{d}E}{\mathrm{d}t} = c^2\Big[(\partial_x u \partial_t u)(x=L) - (\partial_x u \partial_t u)(x=0)\Big]\,.$$

3. Use part (ii) to prove uniqueness of solutions to the wave equation (1). (7 points)

Question 5

Show that the following backward heat equation is not well posed: (7 points)

$$\begin{aligned}\frac{\partial u}{\partial t} &= -k\frac{\partial^2 u}{\partial x^2}, \quad k>0\\ u(0,t) &= 0\\ u(L,t) &= 0\\ u(x,0) &= f(x)\end{aligned}$$

Hint: You can prove this by showing that if you perturb the initial condition by a small amplitude, the solution changes considerably. For example, you can see what happens to the solution if you replace $f(x)$ by $f(x) + \frac{1}{n}\sin\frac{n\pi x}{L}$. This exercise shows that the backward heat equation is unstable, thereby not well posed.

Question 1

(i) We solve $\Delta u = 0$, $(x,y) \in (0,L_1) \times (0,L_2)$ with $u_x(0,y) = u_x(L_1,y) = u_y(x,0) = 0$, $u_y(x,L_2) = f(x)$

By Divergence thm, $0 = \int_{\Omega} \Delta u \, dA = \int_{\partial\Omega} \frac{\partial u}{\partial n} ds$. All boundary fluxes are 0 expect at $y = L_2$, so $0 = \int_0^{L_1} f(x)dx$.

$\Rightarrow$ we need $\boxed{\int_0^{L_1} f(x) dx = 0}$

(ii) From notes, $\phi_n(x) = \cos\left(\frac{n\pi x}{L_1}\right)$, $\lambda_n = \left(\frac{n\pi}{L_1}\right)^2$ with $f(x) = \frac{f_0}{2} + \sum_{n=1}^{\infty} f_n \phi_n(x)$, $f_n = \frac{2}{L_1} \int_0^{L_1} f(x)\phi_n(x) dx$

let $u_n(x,y) = Y_n(y)\phi_n(x) \Rightarrow Y_n'' - \lambda_n Y_n = 0$ then since $u_y(x,0) = 0$, need $Y_n'(0) = 0 \Rightarrow Y_n(y) = A_n \cosh(\sqrt{\lambda_n} y)$, therefore $u(x,y) = C + \sum_{n=1}^{\infty} A_n \cosh(\sqrt{\lambda_n} y)\phi_n(x)$

Then because $u_y(x,L_2) = f(x)$, $\sum_{n=1}^{\infty} A_n \sqrt{\lambda_n} \sinh(\sqrt{\lambda_n} L_2)\phi_n(x) = \frac{f_0}{2} + \sum_{n=1}^{\infty} f_n \phi_n(x) \Rightarrow f_0 = 0$ which shows $\int_0^{L_1} f(x)dx = 0$, which shows (i).

For $n \geq 1$ $A_n = \frac{f_n}{\sqrt{\lambda_n} \sinh(\sqrt{\lambda_n} L_2)}$

$\therefore$ $\boxed{u(x,y) = C + \sum_{n=1}^{\infty} \frac{f_n \cosh(\sqrt{\lambda_n} y)}{\sqrt{\lambda_n} \sinh(\sqrt{\lambda_n} L_2)} \phi_n(x)}$ where $\phi_n(x) = \cos\left(\frac{n\pi x}{L_1}\right)$, $\lambda_n = \left(\frac{n\pi}{L_1}\right)^2$

(iii) If u is the steady state of $v_t = k\Delta v$, $v(x,y,0) = g(x,y) \Rightarrow$ total heat conserved is $\int_0^{L_2}\int_0^{L_1} u(x,y)\,dx\,dy = \int_0^{L_2}\int_0^{L_1} g(x,y)\,dx\,dy$.

Since cosine integrates to 0 in x, only the constant C contributes $\Rightarrow C L_1 L_2 = \int_0^{L_2}\int_0^{L_1} g(x,y)\,dx\,dy$

$\Rightarrow$ $\boxed{C = \frac{1}{L_1 L_2} \int_0^{L_2}\int_0^{L_1} g(x,y)\,dx\,dy}$

Question 2

$\Delta u = 0$, $r > R$ with $u(R,\theta) = \log 2 + 4\cos(3\theta)$

In polar, $\Delta u = u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta}$. The general bounded exterior solution is $u(r,\theta) = A_0 + \sum_{n=1}^{\infty} \left(\frac{R}{r}\right)^n (a_n \cos(n\theta) + b_n \sin(n\theta))$.

We only use decaying powers r^{-n} since the solution is outside the disk and should remain bounded as $r \to \infty$.

At $r = R$, $u(R,\theta) = A_0 + \sum_{n=1}^{\infty} (a_n \cos(n\theta) + b_n \sin(n\theta))$

Match the general solution with $u(R,\theta)$, we get $A_0 = \log 2$, $a_3 = 4 \Rightarrow$ $\boxed{u(r,\theta) = \log 2 + 4\left(\frac{R}{r}\right)^3 \cos(3\theta)}$

Question 3

solve $\Delta u = 0$, $a < r < b$ with $u_r(a,\theta) = 0$, $u(b,\theta) = g(\theta)$

In polar, $\Delta u = u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta}$ with $g(\theta) = \frac{g_0}{2} + \sum_{n=1}^{\infty} (g_n \cos(n\theta) + h_n \sin(n\theta))$ where $g_n = \frac{1}{\pi}\int_0^{2\pi} g(\theta)\cos(n\theta)d\theta$, $h_n = \frac{1}{\pi}\int_0^{2\pi} g(\theta)\sin(n\theta)d\theta$, $g_0 = \frac{1}{\pi}\int_0^{2\pi} g(\theta)d\theta$

The general sol is $u(r,\theta) = A_0 + B_0 \log r + \sum_{n=1}^{\infty} (A_n r^n + B_n r^{-n})\cos(n\theta) + \sum_{n=1}^{\infty} (C_n r^n + D_n r^{-n})\sin(n\theta)$

Since $u_r(r,\theta) = \frac{B_0}{r} + \sum_{n=1}^{\infty} n(A_n r^{n-1} - B_n r^{-n-1})\cos(n\theta) + \sum_{n=1}^{\infty} n(C_n r^{n-1} - D_n r^{-n-1})\sin(n\theta) \Rightarrow B_0 = 0$ and $A_n a^{n-1} = B_n a^{-n-1}$, $C_n a^{n-1} = D_n a^{-n-1} \Rightarrow B_n = A_n a^{2n}$, $D_n = C_n a^{2n}$

so $u(r,\theta) = A_0 + \sum_{n=1}^{\infty} A_n (r^n + a^{2n} r^{-n})\cos(n\theta) + \sum_{n=1}^{\infty} C_n (r^n + a^{2n} r^{-n})\sin(n\theta)$, then $u(b,\theta) = g(\theta) \Rightarrow A_0 = \frac{g_0}{2}$ and

$A_n(b^n + a^{2n}b^{-n}) = g_n$ & $C_n(b^n + a^{2n}b^{-n}) = h_n \Rightarrow A_n = \frac{g_n}{b^n + a^{2n}b^{-n}}$, $C_n = \frac{h_n}{b^n + a^{2n}b^{-n}}$

$\Rightarrow u(r,\theta) = \frac{g_0}{2} + \sum_{n=1}^{\infty} \frac{g_n(r^n + a^{2n}r^{-n})}{b^n + a^{2n}b^{-n}}\cos(n\theta) + \sum_{n=1}^{\infty} \frac{h_n(r^n + a^{2n}r^{-n})}{b^n + a^{2n}b^{-n}}\sin(n\theta)$ where $g_n = \frac{1}{\pi}\int_0^{2\pi} g(\theta)\cos(n\theta)d\theta$, $h_n = \frac{1}{\pi}\int_0^{2\pi} g(\theta)\sin(n\theta)d\theta$

Question 4

solve $u_{tt} = c^2 u_{xx}$ $u(0,t) = u(L,t) = 0$

(i) from seperation of variables $u(x,t) = \sum_{n=1}^{\infty} \left[A_n \cos\left(\frac{n\pi ct}{L}\right) + B_n \sin\left(\frac{n\pi ct}{L}\right)\right] \sin\left(\frac{n\pi x}{L}\right)$. With $\begin{cases} \sin\alpha\cos\beta = \frac{1}{2}(\sin(\alpha+\beta) + \sin(\alpha-\beta)) \\ \sin\alpha\sin\beta = \frac{1}{2}(\cos(\alpha-\beta) - \cos(\alpha+\beta)) \end{cases}$

we can get $u(x,t) = \phi(x+ct) + \psi(x-ct)$ where $\phi(s) = \sum_{n=1}^{\infty}\left[\frac{A_n}{2}\sin\left(\frac{n\pi s}{L}\right) - \frac{B_n}{2}\cos\left(\frac{n\pi s}{L}\right)\right]$, $\psi(s) = \sum_{n=1}^{\infty}\left[\frac{A_n}{2}\sin\left(\frac{n\pi s}{L}\right) + \frac{B_n}{2}\cos\left(\frac{n\pi s}{L}\right)\right]$ as desired.

(ii) take $E(t) = \frac{1}{2}\int_0^L |u_t|^2 dx + \frac{c^2}{2}\int_0^L |u_x|^2 dx \Rightarrow \frac{dE}{dt} = c^2\int_0^L u_t u_{xx} dx + c^2 \int_0^L u_x u_{xt} dx$

integrating by parts: $\int_0^L u_t u_{xx} dx = [u_t u_x]_0^L - \int_0^L u_{tx} u_x dx \Rightarrow \frac{dE}{dt} = c^2[u_t u_x]_0^L - c^2\int_0^L u_{tx} u_x dx \Rightarrow \frac{dE}{dt} = c^2[u_t u_x]_0^L + \left(-c^2\int_0^L u_{tx} u_x dx + c^2\int_0^L u_x u_{xt} dx\right)$

$\Rightarrow$ $\boxed{\frac{dE}{dt} = c^2\left[u_x u_t(L,t) - u_x u_t(0,t)\right]}$

(iii) suppose u, v are two solutions with the same BC and let $w = u - v \Rightarrow w_{tt} = c^2 w_{xx}$ and $w(0,t) = w(L,t) = 0$ and $w(x,0) = 0$, $w_t(x,0) = 0$ and the BCs $w_t(0,t) = w_t(L,t) = 0$

then $\frac{dE_w}{dt} = c^2[(w_x w_t)(L,t) - (w_x w_t)(0,t)] = 0 \Rightarrow E_w(t)$ is constant. And since $E_w(0) = 0$, then $E_w(t) = 0$ $\forall t$.

Thus $w_t = 0$, $w_x = 0$ so w is constant in x and t. Since $w(0,t) = 0$. $\therefore w \equiv 0 \Rightarrow u = v$ so the solutions are unique.

Question 5

$u_t = -k u_{xx}$, $k > 0$ with $u(0,t) = u(L,t) = 0$, $u(x,0) = f(x) \Rightarrow \phi_n(x) = \sin\left(\frac{n\pi x}{L}\right)$, $\lambda_n = \left(\frac{n\pi}{L}\right)^2$. Then seperate variables: $u(x,t) = A_n \phi_n(x) \Rightarrow u_t = A_n'(t)\phi_n(x)$, $u_{xx} = -\lambda_n A_n(t)\phi_n(x)$

Back into equation: $A_n'(t)\phi_n(x) = k\lambda_n A_n(t)\phi_n(x) \Rightarrow A_n'(t) = k\lambda_n A_n(t)$ which makes $A_n(t) = A_n(0)e^{k\lambda_n t}$

then as hinted, perturb IC by $\frac{1}{n}\sin\left(\frac{n\pi x}{L}\right)$ which $\to 0$ as $n \to \infty$

However the backwards heat equation multiplies the n-th mode by exponential factor $e^{k\lambda_n t}$, thus the amplitude of $A_n(t) \sim \frac{1}{n}e^{k\left(\frac{n\pi}{L}\right)^2 t}$ which $\to \infty$ as $n \to \infty$

Thus small changes in the initial condition can produce arbitrary large changes in the solution.

$\therefore$ the solution is unstable & not well posed.